



MyVenue: Managing terabytes of data from thousands of POS devices with BigQuery

Get started for free

GOOGLE CLOUD RESULTS

- ✓ Seamlessly processes 5,000 records per hour during peaks
- ✓ Supports 20,000 dashboard requests over a 12-hour period
- ✓ Minimizes the risk of performance degradation or outages
- ✓ Generates large sales data extracts to dashboards and reports
- ✓ Reduces loads on existing database by 30% during peak periods

MyVenue used Google Cloud and Big Query to process and analyze data from thousands of POS terminals across North America, the UK, and Asia Pacific and additionally improved its scalability and flexibility, thus complementing the rapid integrations of Firebase.

“To manage the large datasets generated by processing tens of millions of sales transactions across more than 140 venues, deliver real-time performance dashboards to thousands of product users, and leverage the large datasets required to generate long-term trend analysis, we needed an equally reliable, fast, and super-scalable data warehousing solution. That’s why we turned to Google Cloud and BigQuery.”

For every extra second a sports fan or concert-goer waits in line to place their food and beverage orders, queue abandonment increases. At a stadium or arena, this represents millions of dollars in lost opportunity annually.

Launched in 2020, [MyVenue](#) provides a cloud-based point-of-sale (POS) solution to venue food and beverage operators in the sport, entertainment, and hospitality industries. The business services more than 140 venues across North America, the UK, and Asia Pacific. Its solution includes a hardware-agnostic POS software, mobile ordering app, suite catering platform with online ordering portal, self-service kiosk, full inventory management, kitchen display system, and browser-accessed back office with real-time reports, dashboards, and configuration updates. As a full service integrator, MyVenue also offers professional services like installation, POS support, and custom development.

Scott Babolka
Marketing Manager,
MyVenue

“MyVenue’s fast, scalable, and robust point-of-sale solution reduces in-venue queue wait times. This improves the guest experience and helps venue operators maximize sales revenue,” says Scott Babolka, Marketing Manager, MyVenue.

“To manage the large datasets generated by processing tens of millions of sales transactions across more than 140 venues, deliver real-time performance dashboards to thousands of product users, and leverage the large datasets required to generate long-term trend analysis, we needed an equally reliable, fast, and super-scalable data warehousing solution. That’s why we turned to Google Cloud and BigQuery.”



Cost, functionality, and ease of use prompt BigQuery selection

Capturing sales transaction data for reporting, analysis, and the provision of real-time metrics to venue food and beverage operators is critical to MyVenue’s value proposition. These metrics enable operators to make adjustments over the course of an event to maximize customer spending, track credit card processing times, monitor latency, and view utilization of the MyVenue POS solution at a venue.

However, as MyVenue’s customer portfolio grew, the demand for large-scale reports and live dashboards increased, as did the strain on database and data

“BigQuery was miles ahead of the competition from a cost and support perspective.

Based on our usage and data segmentation, we were able to come up with a highly cost-effective methodology that met our need for real-time data and to

infrastructure performance. “We understood that our growth trajectory could create future challenges if left unaddressed,” said David Ialeggio, Chief Technology Officer, MyVenue. “If we didn’t meet rising demand, there would be an increased potential for longer data lead times and outages, which in-turn would impact the user experience.”

**query terabytes
of data at a rapid
rate.”**

David Ialeggio
Chief Technology
Officer, MyVenue

With MyVenue’s expertise residing in developing functional and feature-rich solutions rather than building infrastructure, the business began reviewing vendors and products. They needed a provider that could deliver a scalable, workable solution, and produce data on demand without lag, outages, or limitations. Its solution evaluation process was simplified considerably by positive experiences with [Firebase Realtime Database](#) and [Cloud Functions for Firebase](#). These Google Cloud products enable MyVenue to implement third-party integrations and seamlessly push changes, configurations, and data to tens of thousands of POS devices operating in intense environments with patchy connectivity. When connectivity is restored, all data captured from offline sales is synchronized.

After a two-month evaluation, MyVenue selected BigQuery based on cost, functionality, and ease of use. “BigQuery was miles ahead of the competition from a cost and support perspective,” said Ialeggio. “Based on our usage and data segmentation, we were able to come up with a highly cost-effective methodology that met our need for real-time data and to query terabytes of data at a rapid rate.” This would support millions of transactions and venue operators’ growing appetite for data beyond live dashboards and reports, like exports to feed their own analytics systems.

MyVenue then spun up BigQuery instances for each of its development, staging, and production environments to understand how to secure data partitioned by different venues in its multi-tenant environment. The business also needed to work on data inputs and segmentation to support requirements ranging from real-time dashboards to reports based on data collected over months or years.

Finally, the business needed to connect that data and outputs to its back-office systems to support its operational requirements. “Once we had built that framework, we were ready for deployment,” said Ialeggio. “The Google Cloud team provided a series of education sessions that ramped up our knowledge and ensured we completed this exercise in accordance with best practices.”



Extracting data without impact or overheads

“From evaluating the solution, understanding how we could deploy and how we could execute, working with Google Cloud was a very smooth process. With BigQuery, MyVenue is able to provide a best-of-breed point-of-sale solution.”

David Ialeggio
Chief Technology
Officer, MyVenue

With BigQuery, MyVenue extracts data from a range of sources without incurring any overhead from its internal team or experiencing any impact on its infrastructure. “It’s serverless, it’s scalable, handles the workloads, and processes large volumes of data very quickly. This enables us to focus on what we do best, which is to create functionality and product-rich solutions,” said Ialeggio.

With MyVenue’s incumbent database now used primarily for configuration tasks, the load it needs to manage has fallen by 30 percent. During demand peaks, 150,000 records are batched per hour and streamed into the cloud data warehouse in near real-time, while dashboard requests execute at up to 20,000 hits over a 12-hour period.

Venue operators can now extend reports beyond single-day or day-to-day timeframes to months and years without challenging system performance or availability. “After starting with real-time dashboards, we added the ability to generate sales extracts to

dashboards and reports,” said Ialeggio. “This could involve up to 17 million sales records from a single venue alone over a 12-month period, extracted in a nested structure from a browser interface. Helping us provide exact functionality of this power demonstrates the value of BigQuery in enriching the service we provide to customers.”

The MyVenue enterprise-level dashboard powered by BigQuery now displays detailed data trends, and issues ‘hot spots’ around a venue. This allows venue operators to be more efficient and maximize spend from customers by understanding where queues are, where latencies are, and what locations are underutilized and can be bolstered to maximize dollar spend during an event. For example, they can quickly identify locations where queues are building and open up additional terminals or make available a mobile ordering option on the fly to fast-track service. “The feedback we receive from venue operators is that the real-time information we provide enables them to effectively respond to large crowds and considerable movement around food and beverage outlets during major events,” said Ialeggio.

MyVenue is also impressed with the support it received from Google Cloud over its BigQuery deployment journey. “From evaluating the solution, understanding how we could deploy and how we could execute, working with Google Cloud was a very smooth process,” concluded Ialeggio. “With BigQuery, MyVenue is able to provide a best-of-breed point-of-sale solution.”

MyVenue develops cloud-based point-of-sale (POS) solutions for sports, entertainment, and hospitality venues worldwide.

The fast, scalable, and robust technology reduces queue wait times, improves the fan and patron in-venue experience, and helps venue operators maximize sales revenue.

Industry: Technology

Location: Australia, United States, Canada, United Kingdom

Products: [Google Cloud](#), [BigQuery](#), [Cloud Functions for Firebase](#), [Firebase Realtime Database](#)

Why Google	Products and pricing	Solutions	Resources	Engage
Choosing Google Cloud	Google Cloud pricing	Infrastructure modernization	Google Cloud Affiliate Program	Contact sales
Trust and security	Google Workspace pricing	Databases	Google Cloud documentation	Find a Partner
				Become a Partner

Modern Infrastructure Cloud	See all products	Application modernization	Google Cloud quickstarts	Events
Multicloud		Smart analytics	Google Cloud Marketplace	Podcasts
Global infrastructure		Artificial Intelligence	Learn about cloud computing	Developer Center
Customers and case studies		Security	Support	Press Corner
Analyst reports		Productivity & work transformation	Code samples	Google Cloud on YouTube
Whitepapers		Industry solutions	Cloud Architecture Center	Google Cloud Tech on YouTube
Blog		DevOps solutions	Training	Follow on X
		Small business solutions	Certifications	Join User Research
		See all solutions	Google for Developers	We're hiring. Join Google Cloud!
			Google Cloud for Startups	Google Cloud Community
			System status	
			Release Notes	

Sign up for the Google Cloud newsletter

Subscribe